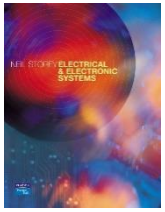
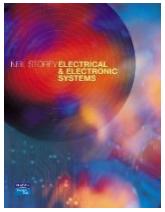


Chapter 7

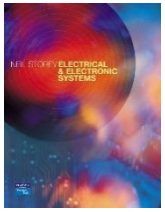
Control and Feedback

- Introduction
- Open-loop and Closed-loop Systems
- Automatic Control Systems
- Feedback Systems
- Negative Feedback
- The Effects of Negative Feedback
- Negative Feedback – A Summary



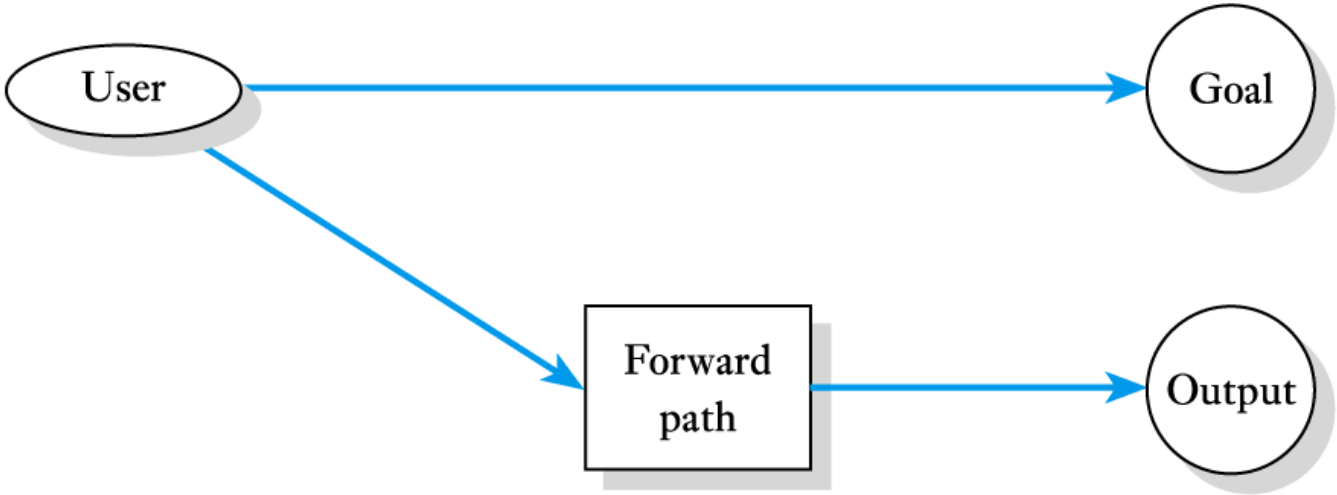
Introduction

- **control** is one of the basic functions performed by many systems
  - often involves **regulation** or **command**
- Invariably, the goal is to determine the value or state of some physical quantity
  - and often to maintain it at that value, despite variations in the system or the environment



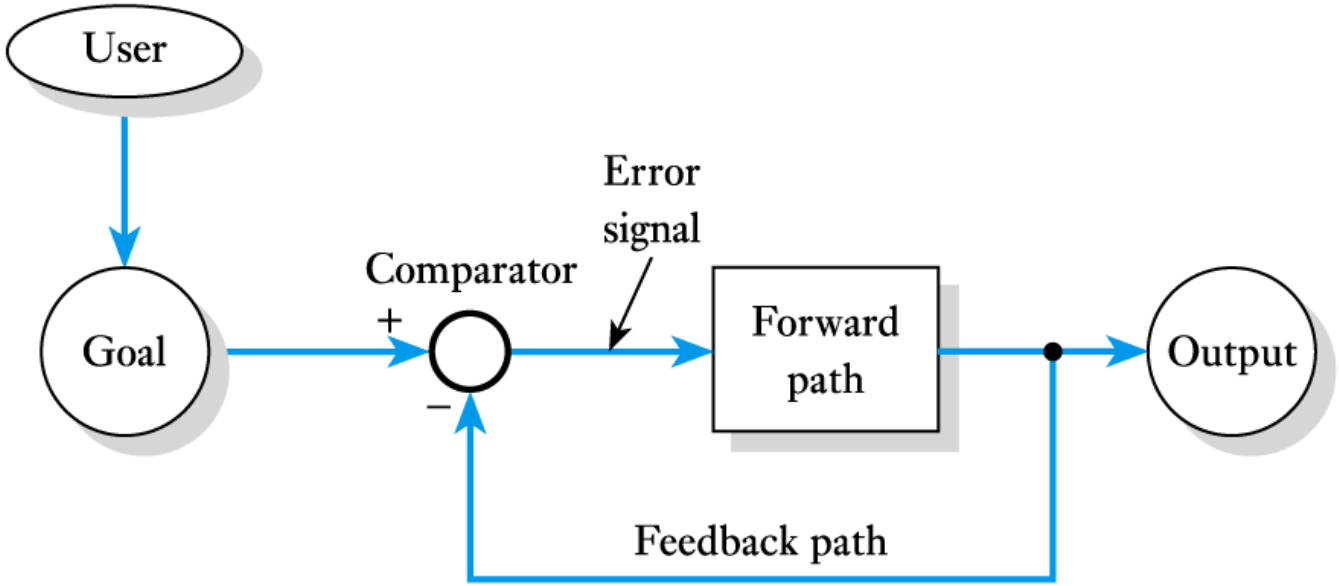
Open-loop and Closed-loop Systems

- Simple control is often **open-loop**
  - user has a **goal** and selects an input to a system to try to achieve this

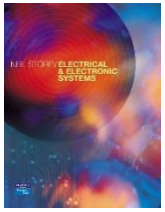


(a) An open-loop system

- More sophisticated arrangements are **closed-loop**
  - user inputs the goal to the system

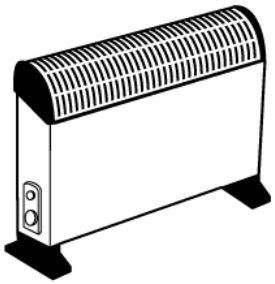


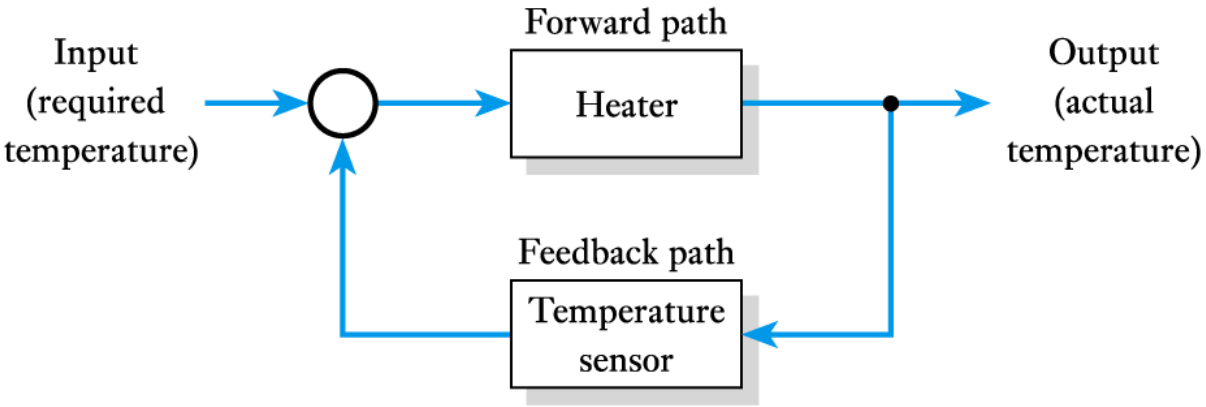
(b) A closed-loop system



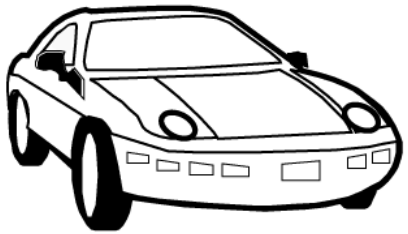
Automatic Control Systems

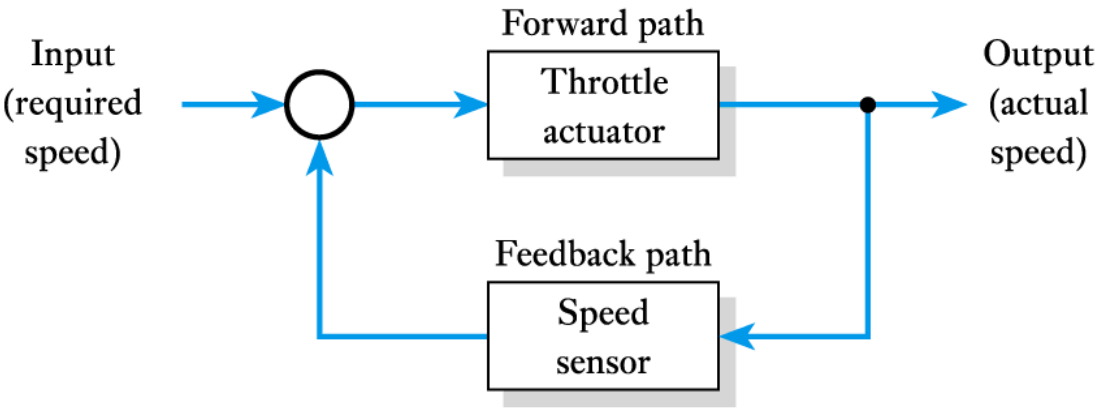
- Examples of automatic control systems:
  - **temperature control using a room heater**



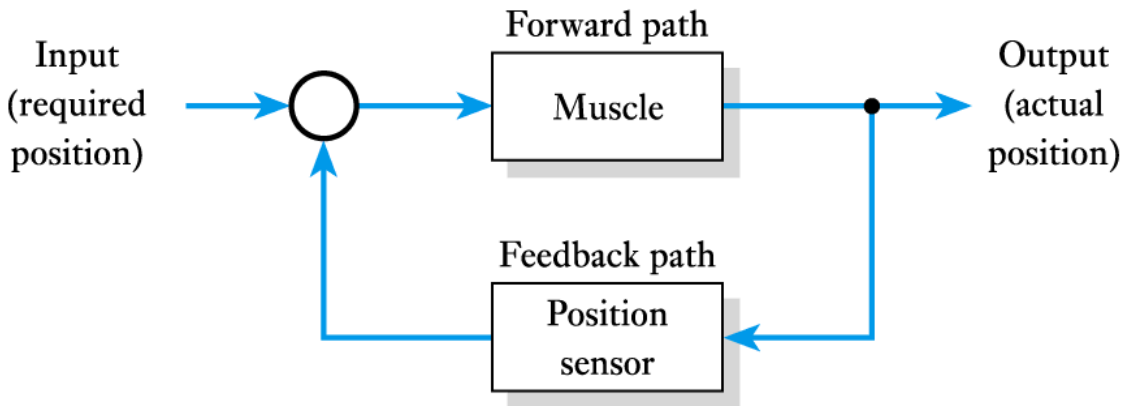


- Examples of automatic control systems:
  - **Cruise control in a car**

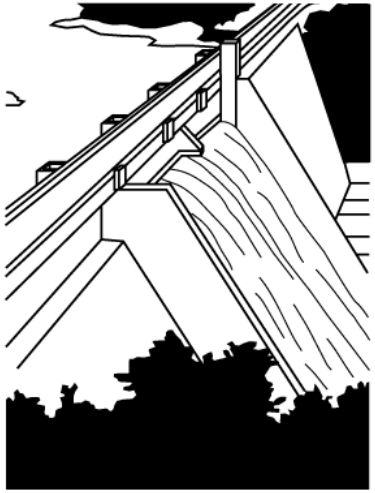


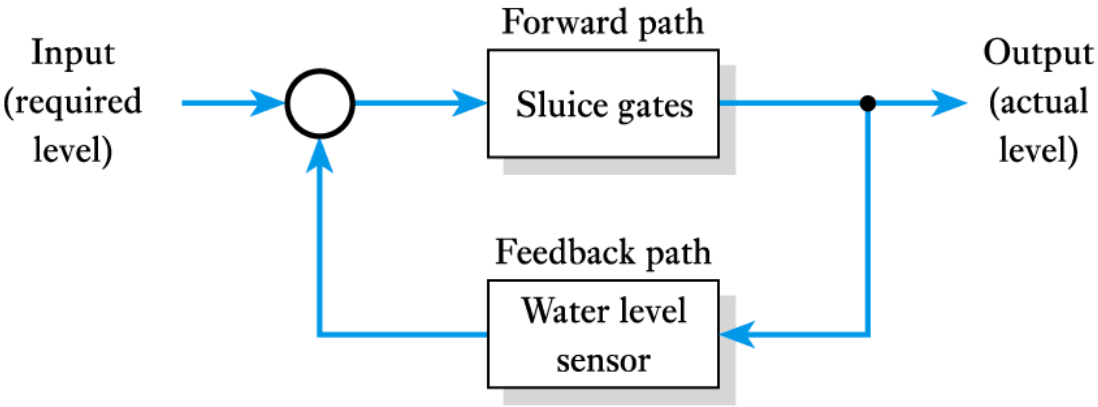


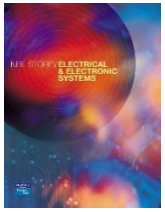
- Examples of automatic control systems:
  - **Position control in a human limb**



- Examples of automatic control systems:
  - **Level control in a dam**

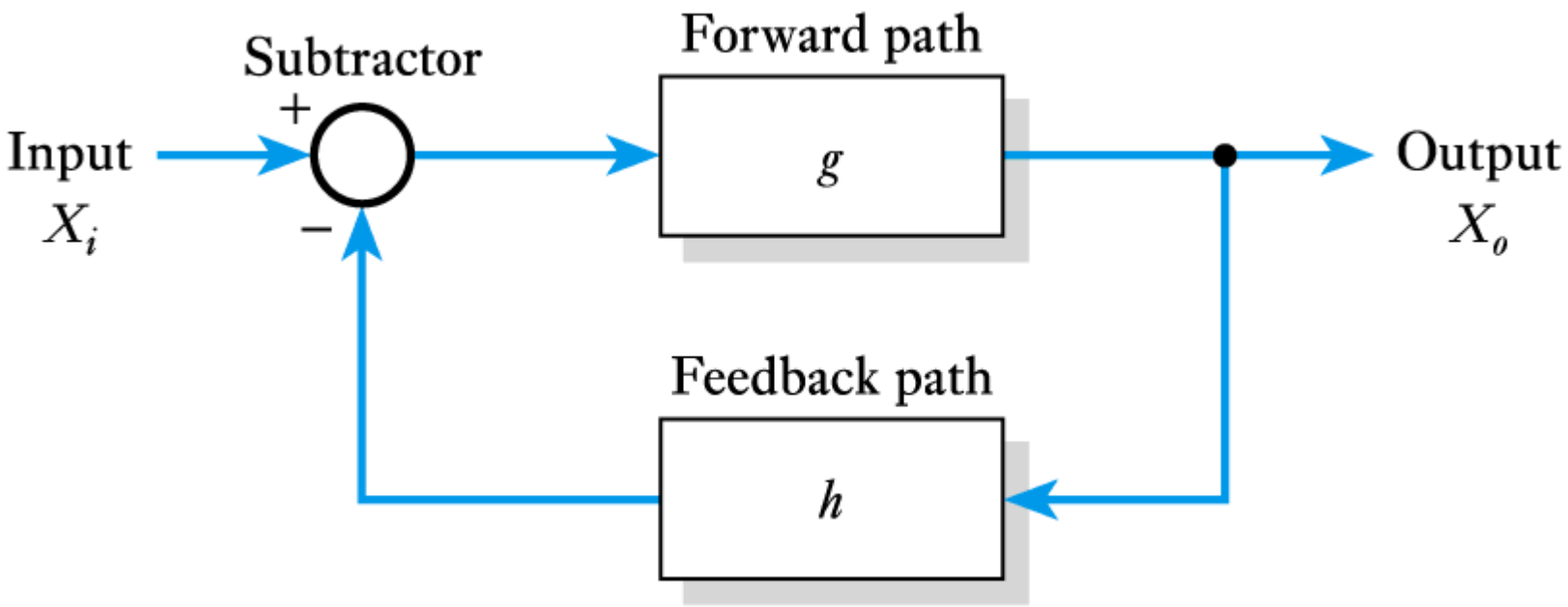




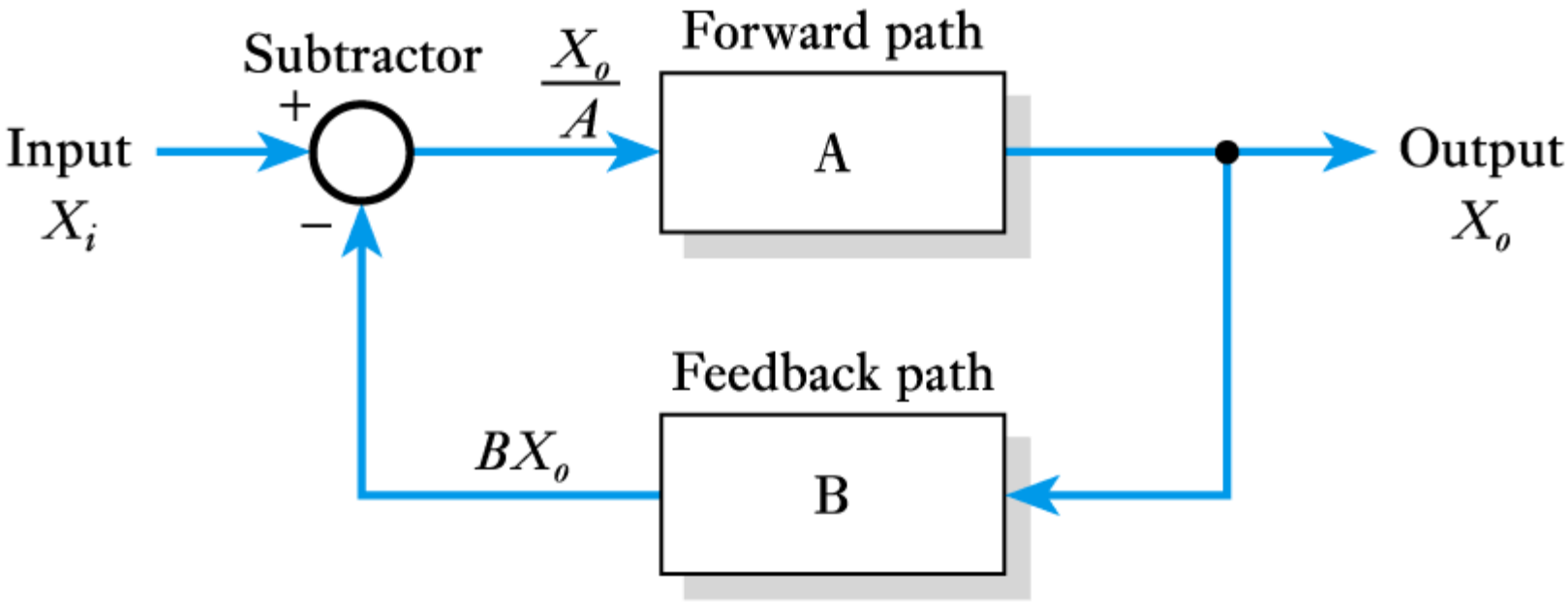


Feedback Systems

- A generalised feedback system



- By inspection of diagram we can add values



$$\frac{X_o}{A} = X_i - BX_o$$

or rearranging

$$\frac{X_o}{X_i} = \frac{A}{1+AB}$$

- Thus

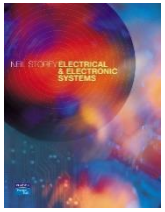
$$\text{Overall gain G} = \frac{X_o}{X_i} = \frac{A}{1 + AB}$$

- This the **transfer function** of the arrangement
- Terminology:
  - *A* is also known as the **open-loop gain**
  - *G* is the overall or **closed-loop gain**

- Effects of the product *AB*
  - If *AB* is negative
    - If *AB* is negative and less than 1, (1 + *AB*) < 1
    - In this situation G > A and we have **positive feedback**
  - If *AB* is positive
    - If *AB* is positive then (1 + *AB*) > 1
    - In this situation G < A and we have **negative feedback**
    - If *AB* is positive *and AB* >>1

$$G = \frac{A}{1 + AB} \approx \frac{A}{AB} = \frac{1}{B}$$

- gain is independent of the gain of the forward path *A*



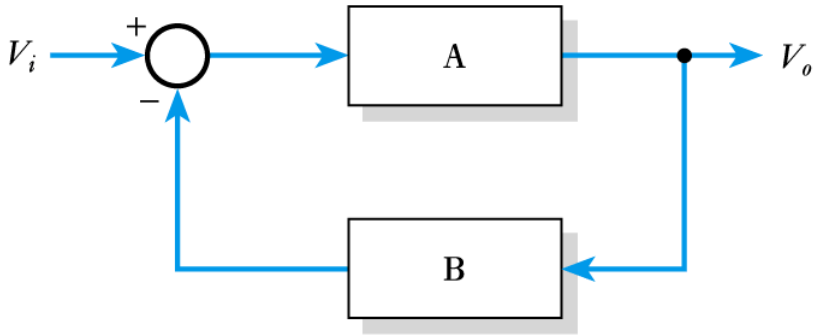
Negative Feedback

- Negative feedback can be applied in many ways
  - X_i and X_o could be temperatures, pressures, etc.
  - here we are mainly interested in voltages and currents
- Particularly important in overcoming **variability**
  - all active devices suffer from variability
    - their gain and other characteristics vary with temperature and between devices
  - we noted above that using negative feedback we can produce an arrangement where the gain is *independent* of the gain of the forward path
    - this gives us a way of overcoming problems of variability

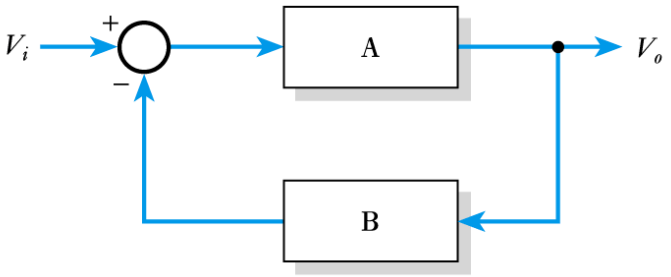
- Consider the following example (**Example 7.1** in text)

  Example: Design an arrangement with a stable voltage gain of 100 using a high-gain active amplifier. Determine the effect on the overall gain of the circuit if the voltage gain of the active amplifier varies from 100,000 to 200,000.

- We will base our design on our standard feedback arrangement



- We will use our active amplifier for *A* and a stable feedback arrangement for *B*

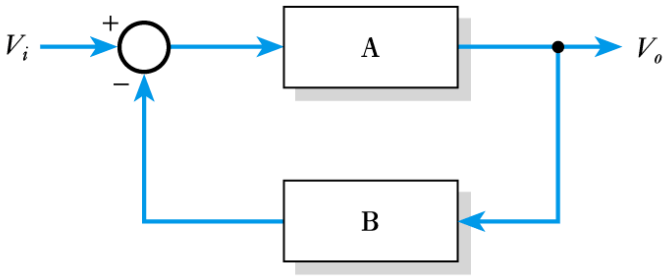


- Since we require an overall gain of 100

$$G = \frac{1}{B}$$

so we will use ***B* = 1/100 or 0.01**

- Now consider the gain of the circuit when the gain of the active amplifier ***A* is 100,000**



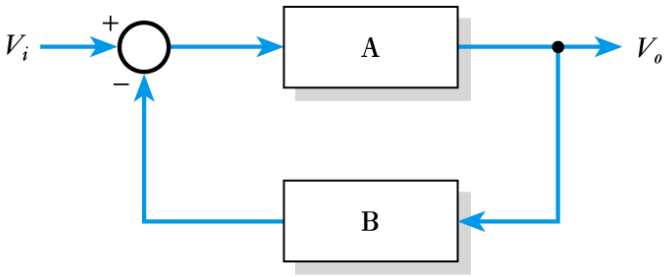
$$G = \frac{A}{1 + AB} = \frac{100\,000}{1 + (100\,000 \times 0.01)}$$

$$= \frac{100\,000}{1 + 1000}$$

$$= 99.90$$

$$\approx \frac{1}{B}$$

- Now consider the gain of the circuit when the gain of the active amplifier ***A* is 200,000**



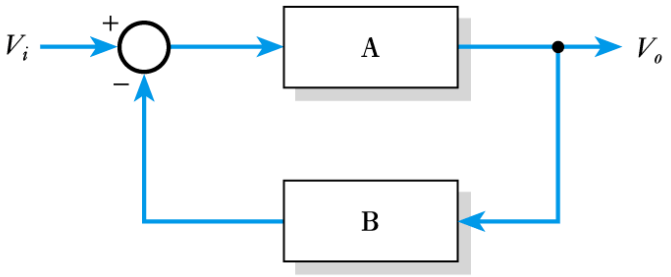
$$G = \frac{A}{1 + AB} = \frac{200\,000}{1 + (200\,000 \times 0.01)}$$

$$= \frac{200\,000}{1 + 2\,000}$$

$$= 99.95$$

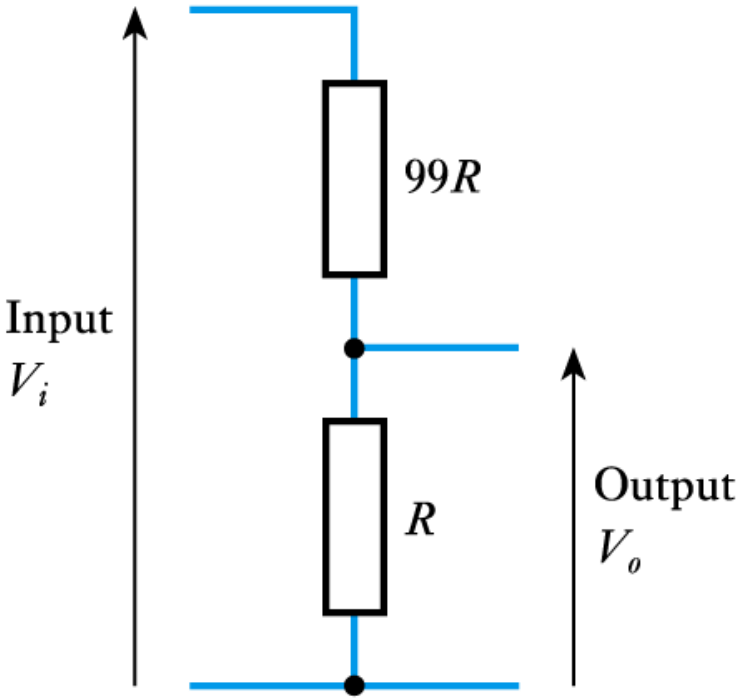
$$\approx \frac{1}{B}$$

- Note that a change in the gain of the active amplifier of 100% causes a change in the overall gain of just 0.05 %



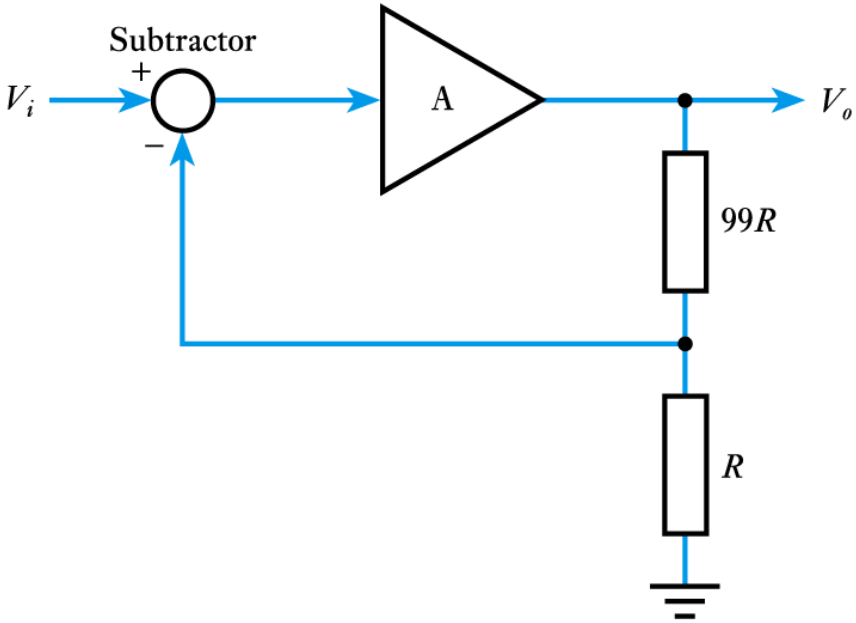
- Thus the use of negative feedback makes the gain largely independent of the gain of the active amplifier
- However, it does require that *B* is stable
  - fortunately, *B* can be based on stable passive components

- Implementing the passive feedback path
  - to get an overall gain of greater than 1 requires a feedback gain *B* of less than 1
  - in the previous example the value of *B* is 0.01
  - this can be achieved using a simple potential divider

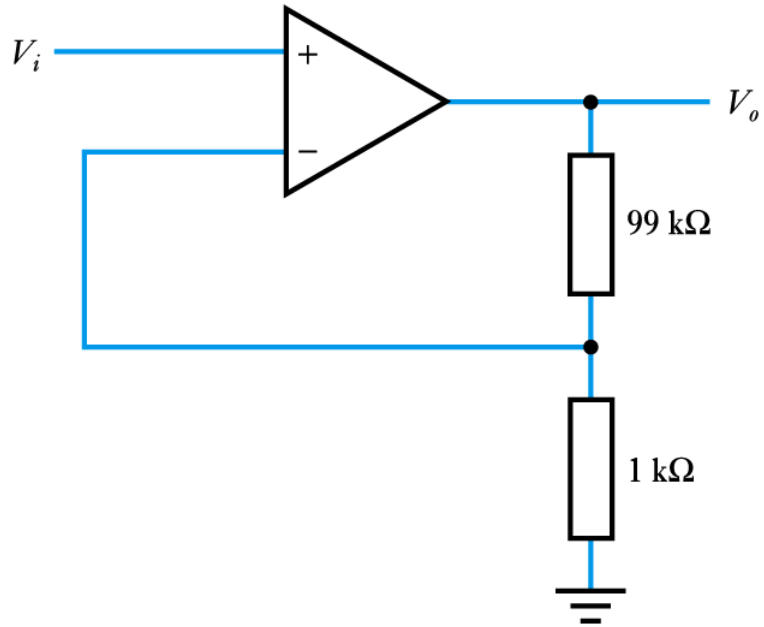


- Thus we can implement our feedback arrangement using an active amplifier and a passive feedback network to produce a stable amplifier
- The arrangement on the right has a gain of 100 …

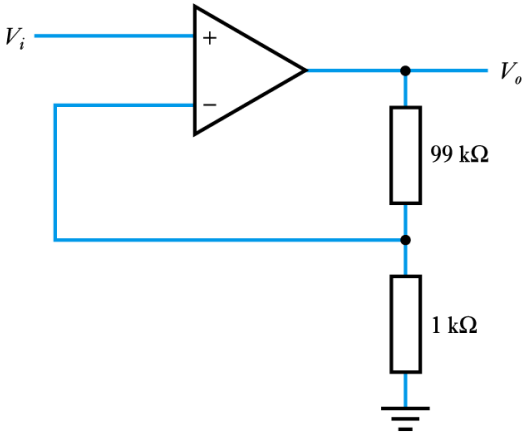
  … but how do we implement the subtractor?

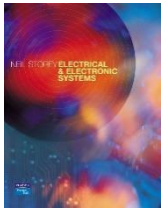


- A differential amplifier is effectively an active amplifier combined with a subtractor. A common form is the **operational amplifier** or **op-amp**
- The arrangement on the right has a gain of 100.



- In this circuit the gain is determined by the passive components and we do not need to know the gain of the op-amp
  - however, earlier we assumed that $AB >> 1$
  - that is, that $A >> 1/B$
  - that is, **open-loop gain >> closed-loop gain**
  - therefore, the gain of the circuit must be much less than the gain of the op-amp
  - see **Example 7.2** in the course text





The Effects of Negative Feedback

- **Effects on Gain**
  - negative feedback produces a gain given by

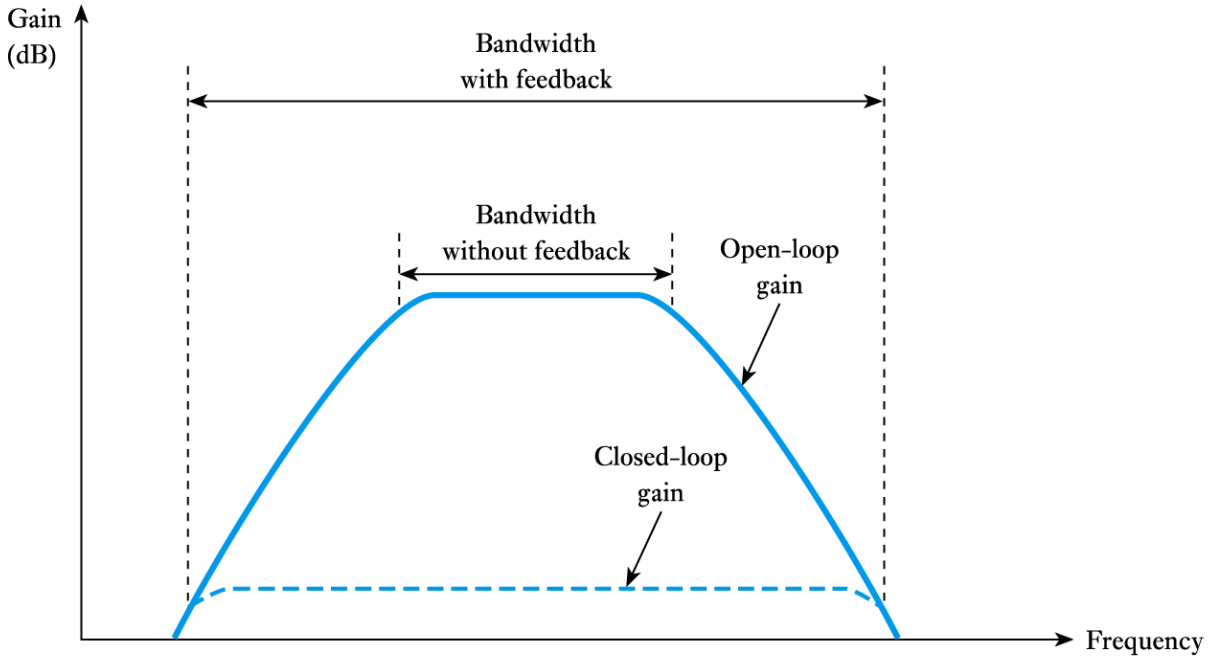
$$G = \frac{A}{1 + AB}$$

  - there, feedback *reduces* the gain by a factor of 1 + *AB*
  - this is the price we pay for the beneficial effects of negative feedback

- **Effects on frequency response**
  - from earlier lectures we know that all amplifiers have a limited frequency response and bandwidth
  - with feedback we make the overall gain largely independent of the gain of the active amplifier
  - this has the effect of increasing the bandwidth, since the gain of the feedback amplifier remains constant as the gain of the active amplifier falls
  - however, when the open-loop gain is no longer much greater than the closed-loop gain the overall gain falls

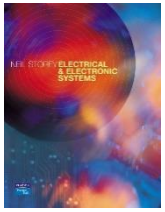
– therefore the bandwidth *increases* as the gain is *reduced* with feedback

– in some cases the **gain x bandwidth = constant**



- **Effects on input and output resistance**
  - negative feedback can either *increase* or *decrease* the input or output resistance depending on how it is used.
    - if the output voltage is fed back this tends to make the output voltage more stable by *decreasing* the output resistance
    - if the output current is fed back this tends to make the output current more stable by *increasing* the output resistance
    - if a voltage related to the output is subtracted from the input voltage this *increases* the input resistance
    - if a current related to the output is subtracted from the input current this *decreases* the input resistance
    - the factor by which the resistance changes is $(1 + AB)$
    - we will apply this to op-amps in a later lecture

- **Effects on distortion and noise**
  - many forms of **distortion** are caused by a non-linear amplitude response
    - that is, the gain varies with the amplitude of the signal
  - since feedback tends to stabilise the gain it also tends to reduce distortion - often by a factor of (1 + *AB*)
  - **noise** produced *within* an amplifier is also reduced by negative feedback – again by a factor of (1 + *AB*)
    - note that noise already corrupting the input signal is *not* reduced in this way – this is amplified along with the signal



Negative Feedback – A Summary

- All negative feedback systems share some properties
  1. They tend to maintain their output independent of variations in the forward path or in the environment
  2. They require a forward path gain that is greater than that which would be necessary to achieve the required output in the absence of feedback
  3. The overall behaviour of the system is determined by the nature of the feedback path
- Unfortunately, negative feedback does have implications for the **stability** of circuits

Key Points

- Feedback is used in almost all automatic control systems
- Feedback can be either negative or positive
- If the gain of the forward path is *A*, the gain of the feedback path is *B* and the feedback is subtracted from the input then

$$G = \frac{A}{1 + AB}$$

- If *AB* is positive and much greater than 1, then $G \approx 1/B$
- Negative feedback can be used to overcome problems of variability within active amplifiers
- Negative feedback can be used to increase bandwidth, and to improve other circuit characteristics.

PID Control for Embedded Systems

Slide Taken From: Richard Ortman and John Bottenberg

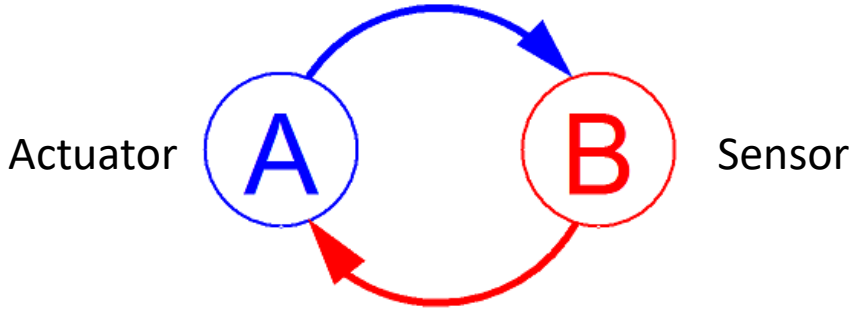
The Problem

- Add/change input to a system

- Did it react how you expected?
  - Could go to fast or slow
  - External environmental factors can play a role (i.e. gravity)

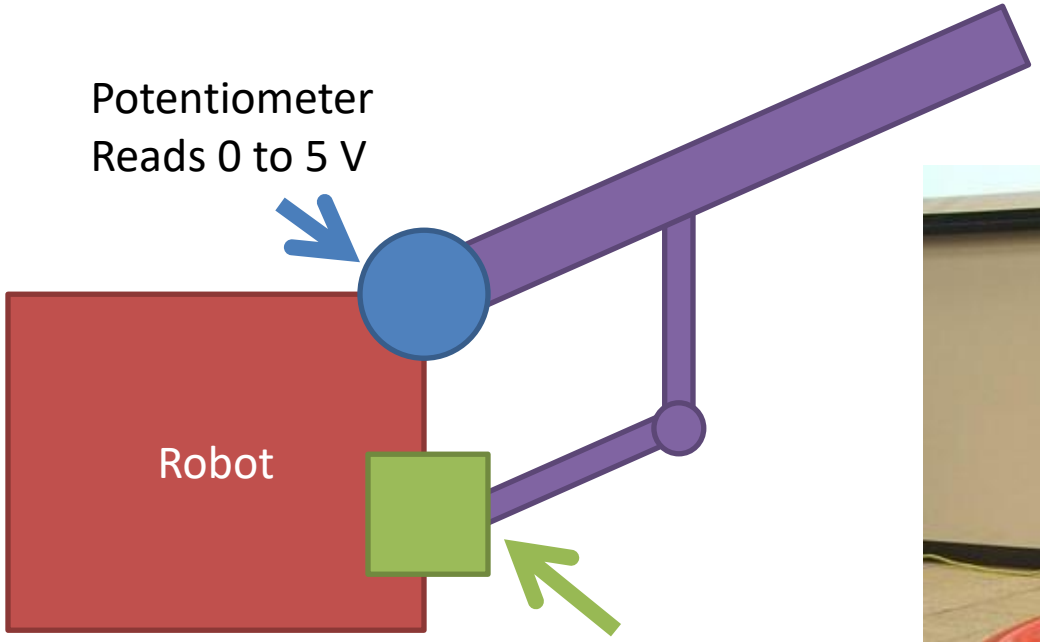
Feedback Control

- Say you have a system controlled by an actuator
- Hook up a sensor that reads the effect of the actuator (NOT the output to the actuator)
- You now have a feedback loop and can use it to control your system!



http://en.wikipedia.org/wiki/File:Simple_Feedback_02.png

Example: Robotic Arm



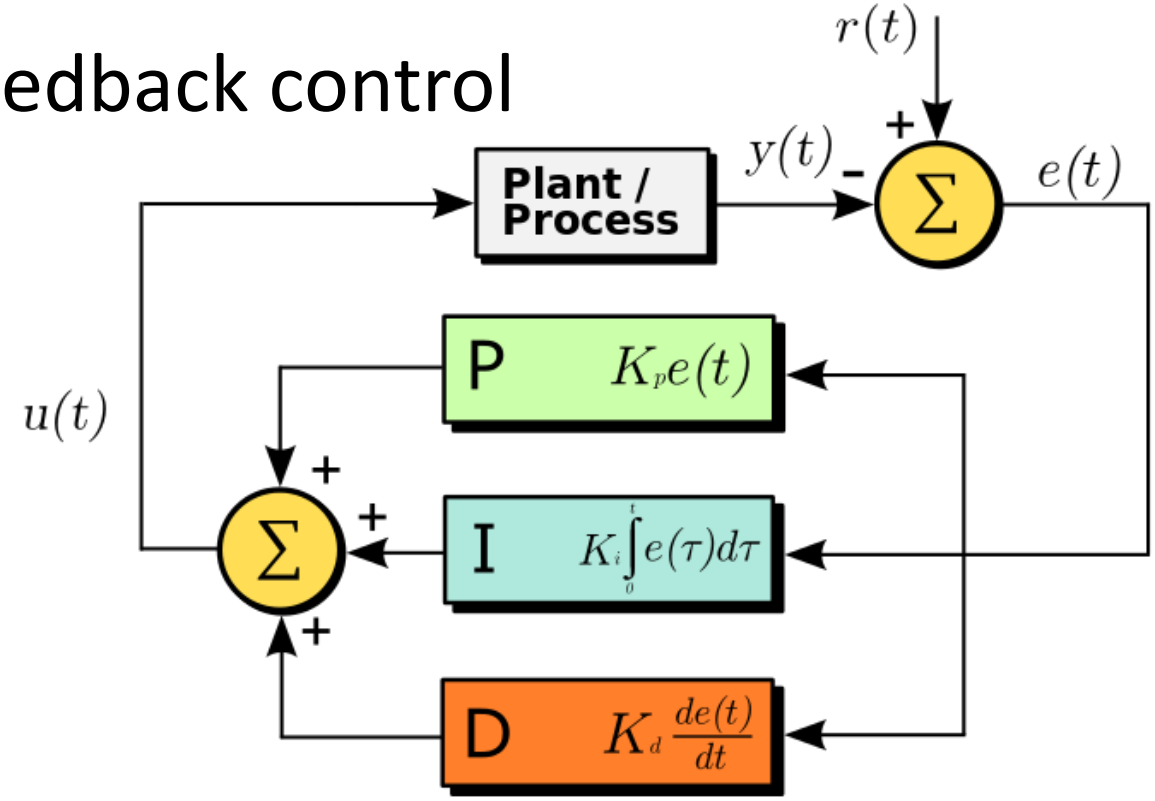
Takes -12 to 12 V
PWM later

http://www.chiefdelphi.com/media/photos/27132

Tell this arm to go to a specified angle

Introduction to PID

- Stands for Proportional, Integral, and Derivative control
- Form of feedback control



http://en.wikipedia.org/wiki/File:PID_en_updated_feedback.svg

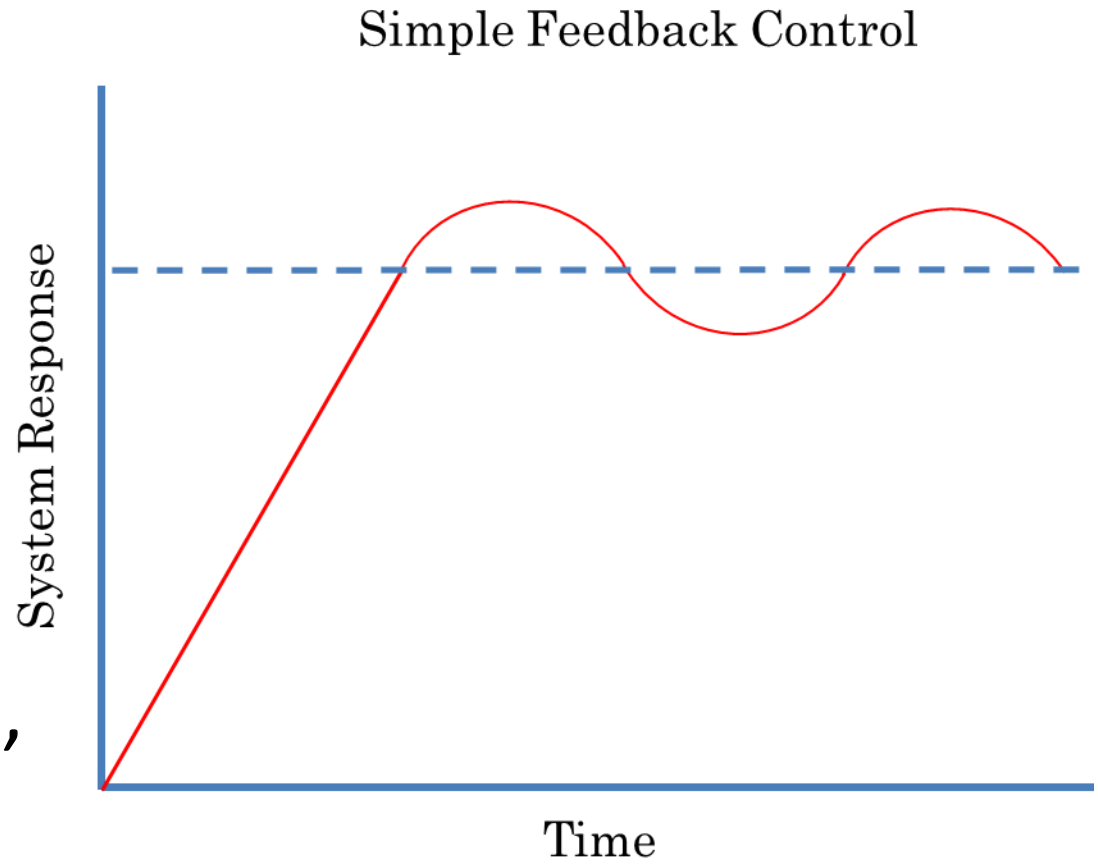
Simple Feedback Control (Bad)

```
double Control (double setpoint, double current) {
      double output;
      if (current < setpoint)
            output = MAX_OUTPUT;
      else
            output = 0;
      return output;
}
```

- Why won't this work in most situations?

Simple Feedback Control Fails

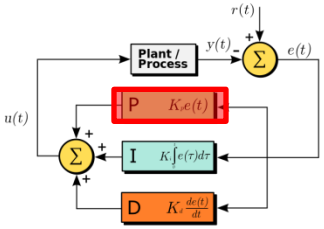
- Moving parts have inertia
- Moving parts have external forces acting upon them (gravity, friction, etc)

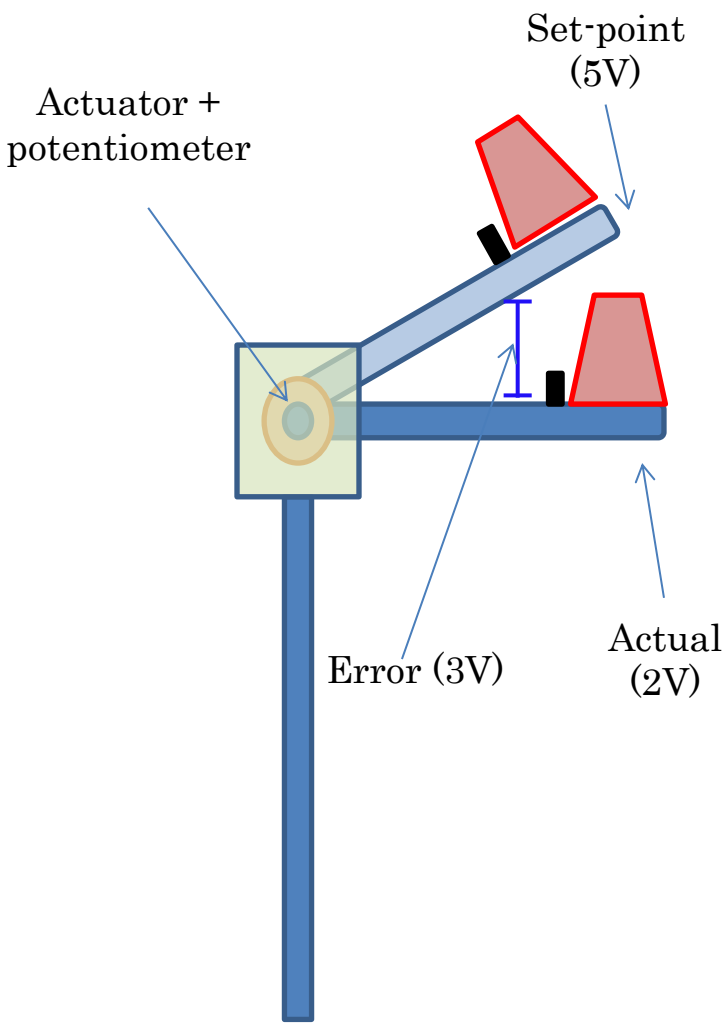


Proportional Control

- Get the error - the distance between the setpoint (desired value) and the actual value
- Multiply it by Kp, the proportional gain
- That's your output!

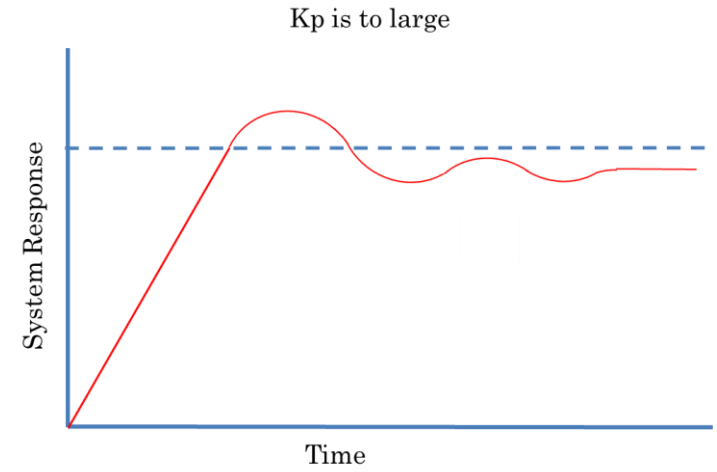
```
double Proportional(double setpoint, double current, double Kp) {
        double error = setpoint - current;
        double P = Kp * error;
        return P;
}
```

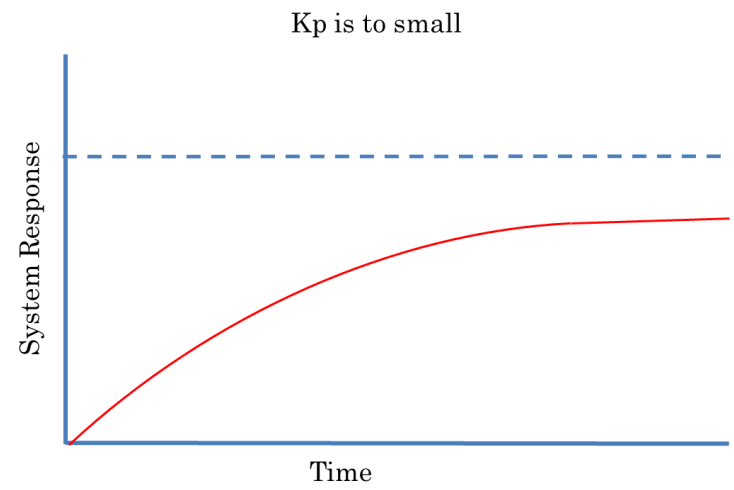



Set-point
(5V)
Actuator +
potentiometer
Error (3V)
Actual
(2V)

Proportional Tuning

- If Kp is too large, the sensor reading will rapidly approach the setpoint, overshoot, then oscillate around it
- If Kp is too small, the sensor reading will approach the setpoint slowly and never reach it





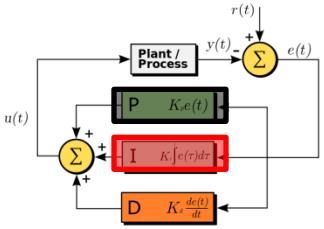
What can go wrong?

- When error nears zero, the output of a P controller also nears zero
- Forces such as gravity and friction can counteract a proportional controller and make it so the setpoint is never reached (steady-state error)
- Increased proportional gain (Kp) only causes jerky movements around the setpoint

Proportional-Integral Control

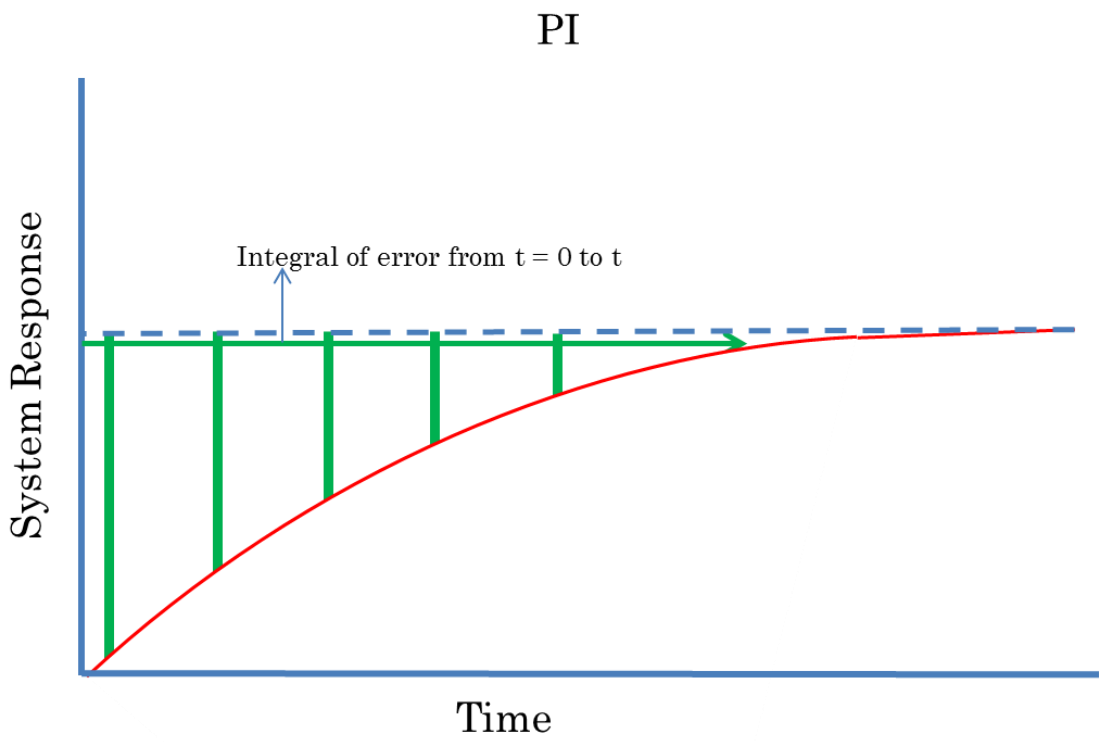
- Accumulate the error as time passes and multiply by the constant Ki. That is your I term. Output the sum of your P and I terms.

```
double PI(double setpoint, double current,
double Kp, double Ki) {
        double error = setpoint - current;
        double P = Kp * error;
        static double accumError = 0;
        accumError += error;
        double I = Ki * accumError;
        return P + I;
}
```



PI controller

- The P term will take care of the large movements
- The I term will take care of any steady-state error not accounted for by the P term



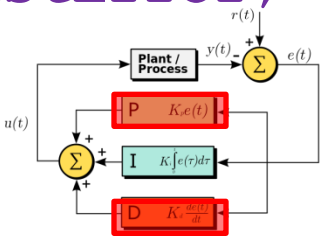
Limits of PI control

- PI control is good for most embedded applications
- Does not take into account how fast the sensor reading is approaching the setpoint
- Wouldn't it be nice to take into account a prediction of future error?

Proportional-Derivative Control

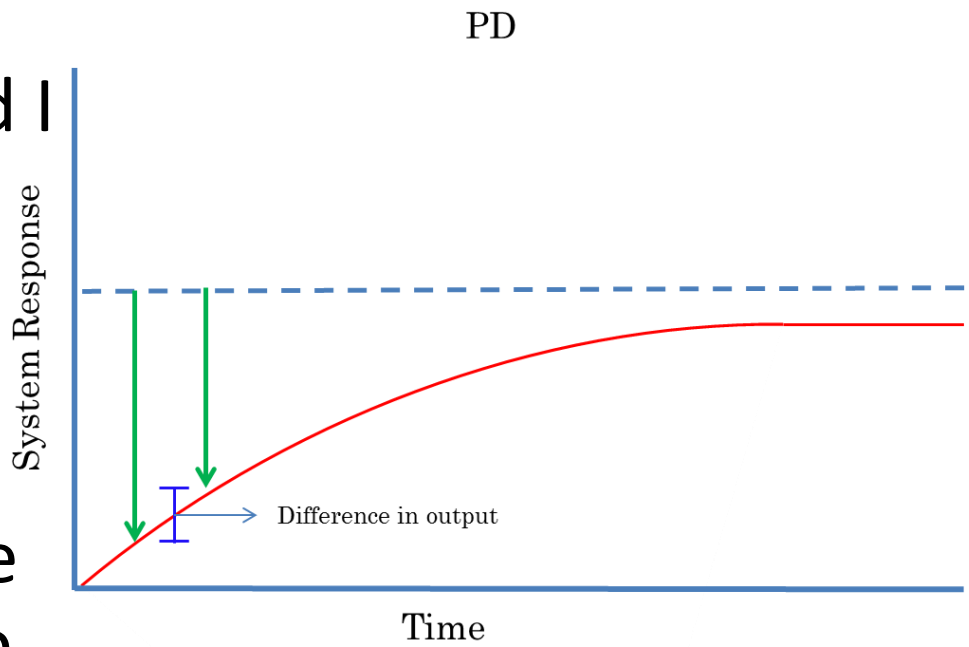
- Find the difference between the current error and the error from the previous timestep and multiply by the constant Kd. That is your D term. Output the sum of your P and D terms.

```
double PD(double setpoint, double current, double
Kp, double Kd) {
        double error = setpoint - current;
        double P = Kp * error;
        static double lastError = 0;
        double errorDiff = error - lastError;
        lastError = error;
        double D = Kd * errorDiff;
        return P + D;
}
```



PD Controller

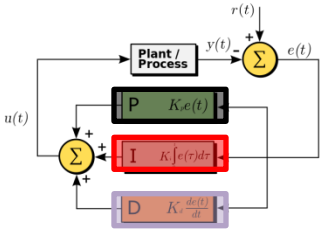
- D may very well stand for "Dampening"
- Counteracts the P and I terms - if system is heading toward setpoint, D term is negative!
- This makes sense: The error is decreasing, so d(error)/dt is negative



PID Control

- Combine P, I and D terms!

```
double PID(double setpoint, double current,
double Kp, double Ki, double Kd) {
      double error = setpoint - current;
      double P = Kp * error;
      static double accumError = 0;
      accumError += error;
      double I = Ki * accumError;
      static double lastError = 0;
      double errorDiff = error - lastError;
      lastError = error;
      double D = Kd * errorDiff;
      return P + I + D;
}
```

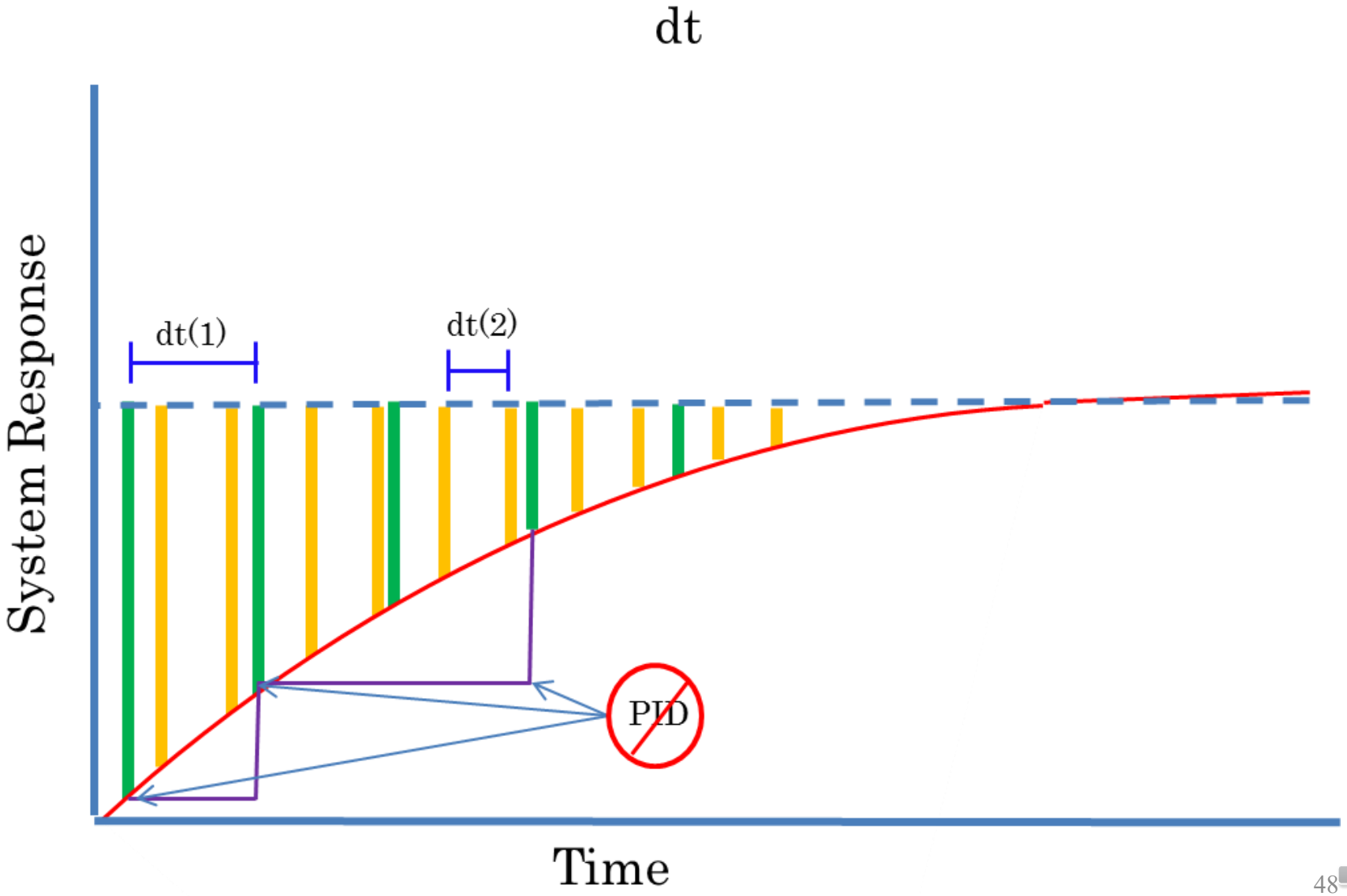


Saturation - Common Mistake

- PID controller can output any value, but actuator has a minimum and maximum input value

```
double saturate(double input, double min,
double max) {
      if (input < min) return min;
      if (input > max) return max;
      return input;
}
// Saturate the output of your PID function
double pid = PID(setpoint, current, Kp, Ki,
Kd);
double output = saturate(pid, min, max);
// Send this output to your actuator!
Actuator = output;
```

Timesteps - Common Mistake



Timesteps - Common Mistake

- Calling a PID control function at different or erratic frequencies results in different behavior
- Regulate this by specifying dt!

```
double PID(double setpoint, double current, double
Kp, double Ki, double Kd, double dt) {
        double error = setpoint - current;
        double P = Kp * error;
        static double accumError = 0;
        accumError += error * dt;
        double I = Ki * accumError;
        static double lastError = 0;
        double errorDiff = (error - lastError) / dt;
        lastError = error;
        double D = Kd * errorDiff;
        return P + I + D;
}
```

PID Tuning

- Start with Kp = 0, Ki = 0, Kd = 0
- Tune P term - System should be at full power unless near the setpoint
- Tune Ki until steady-state error is removed
- Tune Kd to dampen overshoot and improve responsiveness to outside influences
- PI controller is good for most embedded applications, but D term adds stability

PID Applications

- Robotic arm movement (position control)
- Temperature control
- Speed control

Conclusion

- PID uses knowledge about the present, past, and future state of the system, collected by a sensor, to control an actuator
- In PID control, the constants Kp, Ki, and Kd must be tuned for maximum performance